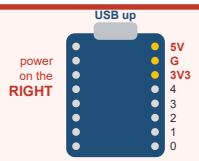
Top view, enlarged for reading

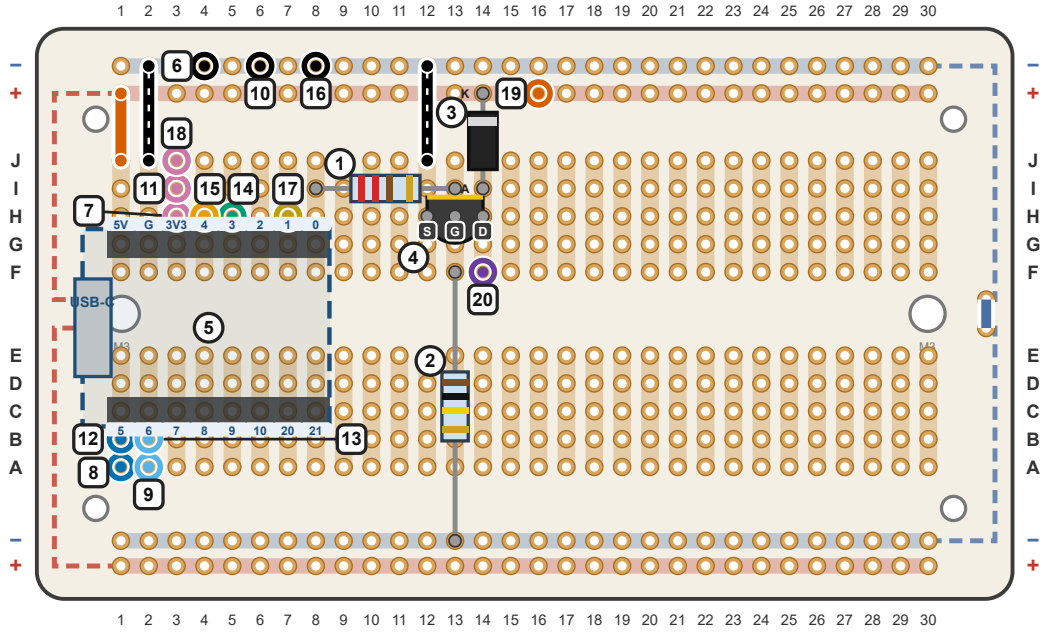
Printer station, C3 POWER-RIGHT — protoboard

TOP VIEW (component side), enlarged, not to scale

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: I8, I13
2. RP1 100 kΩ: F13, bottom − rail at number 13
3. D1 1N4001: top + rail at number 14, I14
4. Q1 BS170: H14, H13, H12
5. U1 ESP32-C3 Super Mini: C1-C8, G1-G8

Device wires

PN532

6. GND: top − rail at number 4
7. 3V3: H3
8. SDA: A1
9. SCL: A2

OLED

10. GND: top − rail at number 6
11. 3V3: I3
12. SDA: B1
13. SCL: B2

encoder

14. CLK: H5
15. DT: H4
16. GND: top − rail at number 8
17. SW: H7
18. 3V3: J3

buzzer

19. 5V: top + rail at number 16
20. BZ−: F14

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

wire end: solder here

copper strip: its holes are joined

(4) a part on the board (see Parts)

(9) a wire of an off-board device (see Device wires)

unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

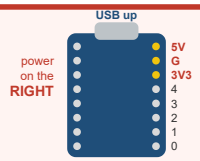
Solder side, enlarged for reading

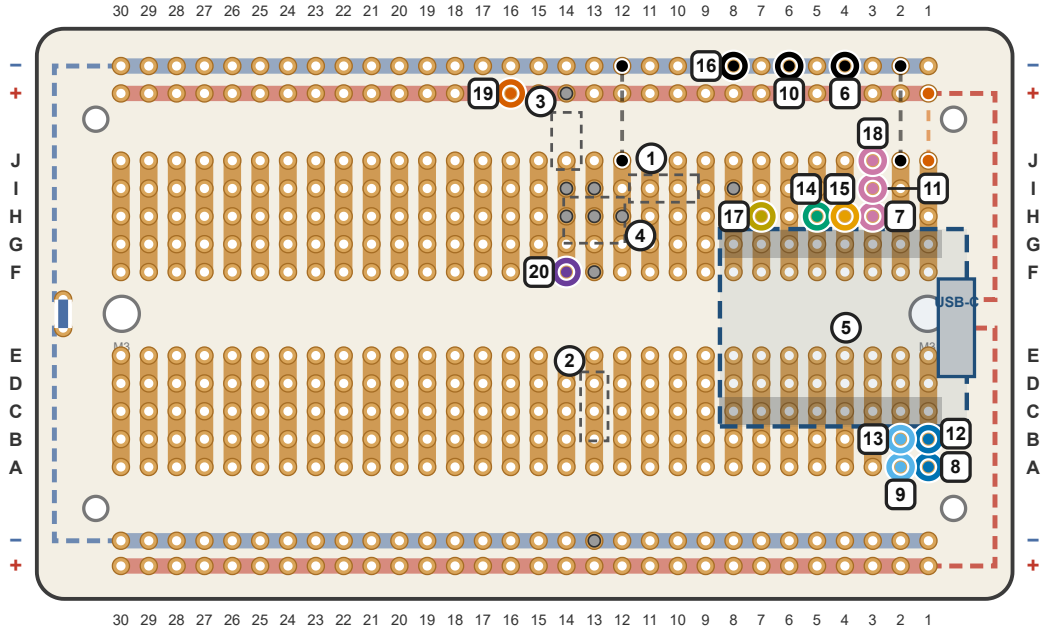
Printer station, C3 POWER-RIGHT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





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OLED

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encoder

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17. SW: H7
18. 3V3: J3

buzzer

19. 5V: top + rail at number 16
20. BZ−: F14

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- [9] a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

Real size (print at 100 %, "actual size"): lay the board on the drawing

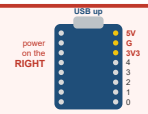
Check the 20 mm bar with a ruler before you rely on the drawing.

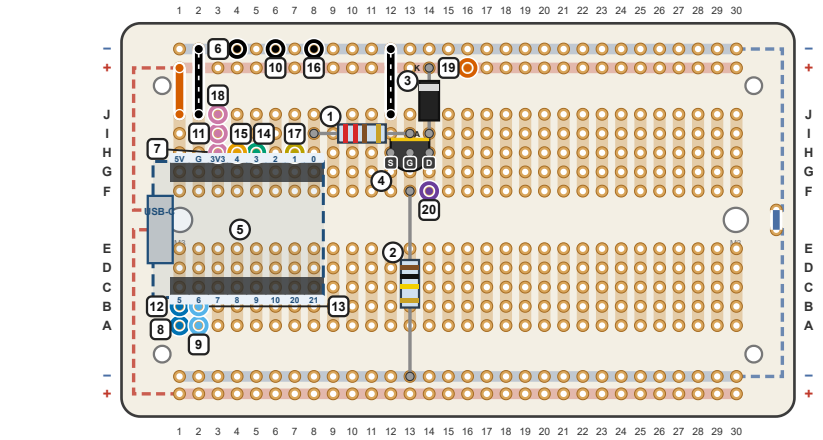
Printer station, C3 POWER-RIGHT — protoboard

TOP VIEW (component side), real size

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





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OLED

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17. SW: H7
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buzzer

19. 5V: top + rail at number 16
20. BZ−: F14

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

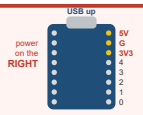
Real size, solder side

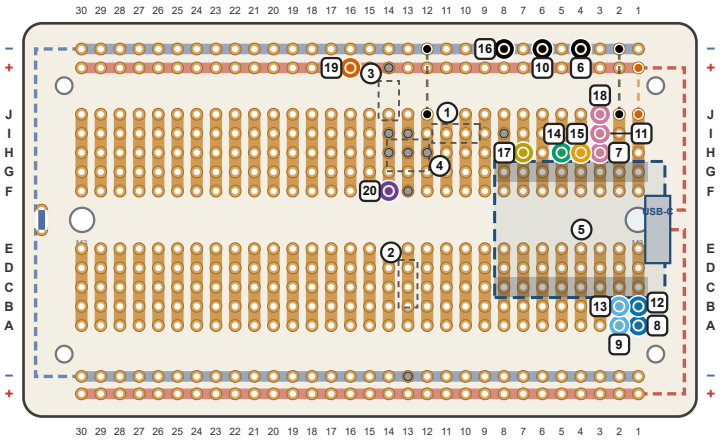
Printer station, C3 POWER-RIGHT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: I8, I13
2. RP1 100 kΩ: F13, bottom − rail at number 13
3. D1 1N4001: top + rail at number 14, I14
4. Q1 BS170: H14, H13, H12
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PN532

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OLED

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encoder

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18. 3V3: J3

buzzer

19. 5V: top + rail at number 16
20. BZ−: F14

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- [9] a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

Build order and hole table

For a C3 POWER-RIGHT module only (USB up: 5V, G, 3V3 at the top of the RIGHT row). A power-left module in this layout puts 5 V on a GPIO pin and can destroy it.

Build order (lowest parts first). The numbers are the callouts on the drawing; wire links carry no number:

- Rail link (−): wire from the right link pad, upper to the right link pad, lower. It joins the top and bottom − rails.
- Wire GND: J2 to top − rail at number 2 (straight, 9 mm).
- Wire GND: J12 to top − rail at number 12 (straight, 9 mm).
- Wire 5V: J1 to top + rail at number 1 (straight, 6 mm).
- Part 1, RG1 (220 Ω): I8 to I13, lying flat.
- Part 2, RP1 (100 kΩ): F13 to bottom − rail at number 13, lying flat.
- Part 3, D1 (1N4001): banded end (K) in top + rail at number 14, other end (A) in I14.
- Part 4, Q1 (BS170): D in H14, G in H13, S in H12, flat face toward the top edge (the J side).
- Part 5, U1 sockets: two 1x8 female headers in C1-C8, G1-G8. Plug U1 in while soldering to keep them square, then take it out.
- Device wires 6-9, PN532: 6 GND in top − rail at number 4, 7 3V3 in H3, 8 SDA in A1, 9 SCL in A2. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 10-13, OLED: 10 GND in top − rail at number 6, 11 3V3 in I3, 12 SDA in B1, 13 SCL in B2. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 14-18, encoder: 14 CLK in H5, 15 DT in H4, 16 GND in top − rail at number 8, 17 SW in H7, 18 3V3 in J3. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 19-20, buzzer: 19 5V in top + rail at number 16, 20 BZ− in F14. Solder each wire from the top, then tie the bundle to the board edge.
- Check with a meter: no short from 5V, 3V3 or SDA/SCL to GND. Then plug U1 in, USB-C toward the board edge as drawn, after checking it is a power-right module.

No.	Item	Holes
	− rail link	right link pad, upper to right link pad, lower
	wire GND	J2 to top − rail at number 2
	wire GND	J12 to top − rail at number 12
	wire 5V	J1 to top + rail at number 1
1	RG1 220 Ω	I8 to I13
2	RP1 100 kΩ	F13 to bottom − rail at number 13
3	D1 1N4001	K top + rail at number 14, A I14
4	Q1 BS170	D H14, G H13, S H12
5	U1 sockets	C1-C8, G1-G8
6-9	PN532 wires	6 GND top − rail at number 4, 7 3V3 H3, 8 SDA A1, 9 SCL A2
10-13	OLED wires	10 GND top − rail at number 6, 11 3V3 I3, 12 SDA B1, 13 SCL B2
14-18	encoder wires	14 CLK H5, 15 DT H4, 16 GND top − rail at number 8, 17 SW H7, 18 3V3 J3
19-20	buzzer wires	19 5V top + rail at number 16, 20 BZ− F14